

MACHINE LEARNING FUNDAMENTALS SERIES

Mutual Information: Fundamentals & Applications

Quantifying Feature Dependency, Non-Linear Relationships, and Information Sharing

Source: StatQuest with Josh Starmer — Machine Learning Fundamentals

1 Introduction & Motivation

In machine learning workflows, feature selection is a critical step aimed at reducing data collection costs and model complexity. The primary goal is to determine how much information each variable (feature) provides about the target variable we wish to predict (e.g., predicting whether a person Loves Troll 2: Yes or No).

- **Why not R^2 or Pearson Correlation?** While R^2 quantifies relationships between variables, it is strictly designed for *continuous* data and only captures *linear* dependencies. In contrast, many real-world datasets contain a mix of discrete and continuous variables (e.g., Likes Popcorn [Yes/No] and Height [Continuous]).
- **The Mutual Information Solution:** Mutual Information (MI) provides a universal, numeric metric to quantify how closely related two variables are, regardless of whether they are discrete or continuous, or whether their relationship is linear or highly non-linear.

2 Joint vs. Marginal Probabilities

Calculating Mutual Information relies on two fundamental probability concepts:

- **Joint Probability $P(X, Y)$:** The probability of two events occurring simultaneously (e.g., the probability that someone *likes popcorn* AND *loves Troll 2*).
- **Marginal Probability $P(X)$ or $P(Y)$:** The probability of a single event occurring independently of other variables, located in the "margins" of a probability table.

2.1 Motivating Example Dataset (5 Observations)

Consider a dataset of 5 individuals evaluated for Likes Popcorn (X) and Loves Troll 2 (Y):

Table 1: Joint and Marginal Probability Table for Likes Popcorn vs. Loves Troll 2

Loves Troll 2 (Y) \ Likes Popcorn (X)	Yes	No	Marginal $P(Y)$
Yes	$3/5 = 0.60$	$1/5 = 0.20$	$4/5 = 0.80$
No	$0/5 = 0.00$	$1/5 = 0.20$	$1/5 = 0.20$
Marginal $P(X)$	$3/5 = 0.60$	$2/5 = 0.40$	$5/5 = 1.00$

3 Mathematical Definition & Sample Calculation

The general equation for Mutual Information between two discrete random variables X and Y is defined as:

$$I(X; Y) = \sum_{x \in X} \sum_{y \in Y} P(x, y) \log \left(\frac{P(x, y)}{P(x)P(y)} \right) \quad (1)$$

Note: The logarithm can use any base. Natural logarithm ($\ln$) is the default in most machine learning implementations.

3.1 Step-by-Step Sample Calculation for Motivating Example

Expanding the double summation across all 4 combinations of $X \in \{\text{Yes}, \text{No}\}$ and $Y \in \{\text{Yes}, \text{No}\}$:

$$\begin{aligned} I(X; Y) = & \underbrace{P(\text{Yes}, \text{Yes}) \ln \left(\frac{P(\text{Yes}, \text{Yes})}{P_X(\text{Yes})P_Y(\text{Yes})} \right)}_{\text{Term 1: (Yes, Yes)}} + \underbrace{P(\text{Yes}, \text{No}) \ln \left(\frac{P(\text{Yes}, \text{No})}{P_X(\text{Yes})P_Y(\text{No})} \right)}_{\text{Term 2: (Yes, No)}} \\ & + \underbrace{P(\text{No}, \text{Yes}) \ln \left(\frac{P(\text{No}, \text{Yes})}{P_X(\text{No})P_Y(\text{Yes})} \right)}_{\text{Term 3: (No, Yes)}} + \underbrace{P(\text{No}, \text{No}) \ln \left(\frac{P(\text{No}, \text{No})}{P_X(\text{No})P_Y(\text{No})} \right)}_{\text{Term 4: (No, No)}} \end{aligned}$$

Substituting numerical values from Table 1:

- **Term 1:** $\frac{3}{5} \ln \left(\frac{3/5}{(3/5)(4/5)} \right) = 0.6 \times \ln(1.25) \approx 0.1339$
- **Term 2:** $\frac{0}{5} \ln \left(\frac{0/5}{(3/5)(1/5)} \right) = 0 \times \ln(0) = 0.0000$ (See Technical Note below)
- **Term 3:** $\frac{1}{5} \ln \left(\frac{1/5}{(2/5)(4/5)} \right) = 0.2 \times \ln(0.625) \approx -0.0940$
- **Term 4:** $\frac{1}{5} \ln \left(\frac{1/5}{(2/5)(1/5)} \right) = 0.2 \times \ln(2.5) \approx 0.1833$

Summing all terms yields the final Mutual Information:

$$I(\text{Likes Popcorn}; \text{Loves Troll 2}) = 0.1339 + 0 + (-0.0940) + 0.1833 \approx \mathbf{0.22}$$

Super Technical Note: Handling $\log(0) \cdot 0$

In Term 2, because $P(\text{Yes}, \text{No}) = 0$, the expression evaluates to $0 \cdot \log(0)$. Strictly speaking, $\log(0)$ is undefined ($\lim_{x \rightarrow 0^+} \log x = -\infty$). However, applying L'Hôpital's Rule from calculus demonstrates:

$$\lim_{x \rightarrow 0^+} x \ln(x) = \lim_{x \rightarrow 0^+} \frac{\ln(x)}{1/x} = \lim_{x \rightarrow 0^+} \frac{1/x}{-1/x^2} = \lim_{x \rightarrow 0^+} (-x) = 0$$

Therefore, any term where the joint probability $P(x, y) = 0$ evaluates strictly to **0**.

4 Experimental Scenarios & Mutual Information Behavior

To understand how Mutual Information behaves under different data relationships, consider four distinct experimental scenarios:

4.1 Scenario 1: Constant Feature (Zero Variation)

- **Setup:** Likes Popcorn is *Yes* for all observations (never changes).
- **Result:** $I(X; Y) = 0.00$.
- **Core Takeaway:** A feature that never changes contains zero entropy/surprise. Something that never changes cannot provide any information about something that changes.

4.2 Scenario 2: Identical Changes (Perfect Positive Association)

- **Setup:** Both columns change in the exact same way (Likes Popcorn is *Yes* whenever Loves Troll 2 is *Yes*, and *No* whenever *No*).
- **Result:** $I(X; Y) \approx 0.50$ (significantly higher than 0.22).
- **Core Takeaway:** Knowing one column provides complete information about the state of the second column.

4.3 Scenario 3: Opposite Changes (Perfect Negative Association)

- **Setup:** Both columns change in exact opposite ways (Likes Popcorn is *Yes* whenever Loves Troll 2 is *No*, and vice versa).
- **Result:** $I(X; Y) \approx 0.50$ (identical to Scenario 2).
- **Core Takeaway:** Mutual Information measures the *magnitude of shared information/dependency*, regardless of whether the correlation is positive or negative.

4.4 Scenario 4: Continuous Variables (e.g., Height)

- **Setup:** When evaluating a continuous feature like Height, the feature values are binned into a histogram (e.g., 3 bins).
- **Procedure:** Each histogram bin serves as a discrete category, expanding the probability matrix to 2×3 (6 total terms in the summation).
- **Result:** $I(\text{Height}; \text{Loves Troll 2}) = 0.22$. Since both Likes Popcorn and Height yield 0.22, either feature is equally informative for predicting Loves Troll 2.

5 Relationship Between Mutual Information and Entropy

Mutual Information is deeply rooted in Information Theory and shares an intimate mathematical relationship with Entropy:

Information Theory Identities

$$I(X; Y) = H(X) + H(Y) - H(X, Y) \quad (\text{Sum of individual entropies minus joint entropy}) \quad (2)$$

$$I(X; Y) = H(X) - H(X|Y) \quad (\text{Reduction in uncertainty of } X \text{ given } Y) \quad (3)$$

$$I(X; Y) = H(Y) - H(Y|X) \quad (\text{Reduction in uncertainty of } Y \text{ given } X) \quad (4)$$

$$I(X; Y) = D_{\text{KL}}(P(X, Y) \parallel P(X)P(Y)) \quad (\text{Kullback-Leibler divergence to independence}) \quad (5)$$

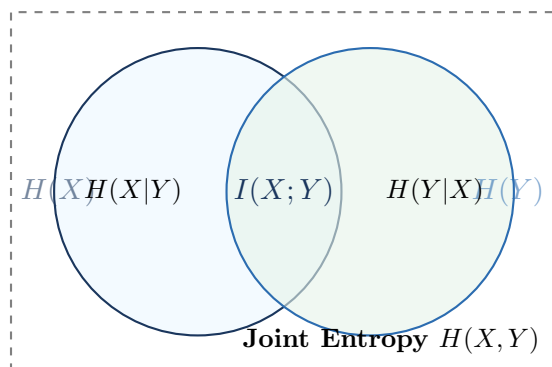


Figure 1: Venn Diagram Representation of Information Theory Concepts. Mutual Information $I(X; Y)$ represents the shared information (intersection) between $H(X)$ and $H(Y)$.

5.1 Key Conceptual Insights

1. **Shared Surprise:** Mutual Information quantifies how much, on average, the "surprise" or variation observed in one variable is shared with the surprise/variation in another variable.
2. **Symmetry:** Unlike conditional entropy, Mutual Information is strictly symmetric: $I(X; Y) = I(Y; X)$.
3. **Non-negativity:** $I(X; Y) \geq 0$, with equality holding if and only if X and Y are completely independent ($P(X, Y) = P(X)P(Y)$).